

Multidomain burden among midlife and older women across seven international ageing cohorts: a harmonized descriptive audit and cautionary profile-stability analysis

Feifan Lu^{1†}, Jun Chen^{2†}, Jizi Shen^{1†}, Li Li^{1*}, Rui Guan^{1*}

^{1*}Department of Obstetrics and Gynecology, Changhai Hospital, Naval Medical University, Shanghai, 200433, Shanghai, China.

²Department of General Surgery, The 989 Hospital of the PLA Joint Service Support Force, No.2 Huaxia West Road, Luoyang, 471031, China.

*Corresponding author(s). E-mail(s): 13764260296@163.com; cngreen785@163.com;

Contributing authors: luff94@163.com; chenjunsmmu@163.com; 664331614@qq.com;

[†]Feifan Lu, Jun Chen and Jizi Shen contributed equally to this work.

Abstract

Background: Women’s health research in later life increasingly needs multidomain measures that capture physical functioning, cognition, affective symptoms and chronic disease burden. Categorical profiles may help communicate heterogeneous burden patterns, but they can be overinterpreted when harmonization, complete-case selection, endpoint coupling and model stability are not explicit. **Methods:** We analysed approved, downloaded, de-identified cohort datasets for women aged 50 years or older in CHARLS, ELSA, HRS, KLoSA, LASI, MHAS and SHARE. The authors cleaned the local cohort files, used harmonized ageing-study variables where available [1] and locally assembled domain inputs when required. Four within-cohort burden domains were oriented so that higher scores indicated worse burden. We treated Gaussian mixture models as descriptive partitions and audited measurement comparability, denominator attrition, LFO functional-change associations, algorithm concordance, formal secondary mortality models, all-sex interaction screens and prediction guardrails. **Results:** The source screen included 79,938 women, of whom 76,293 had

complete four-domain descriptive assignments; the complete-domain 50–64-year stratum ranged from 45.4% to 67.8% across cohorts. Strict-core construction comprised CHARLS, ELSA, HRS and MHAS (29,058 women); SHARE, KLoSA and LASI remained sensitivity or descriptive tiers. All selected full-covariance four-domain Gaussian mixture solutions triggered near-singular covariance diagnostics, and cross-method agreement was limited in several cohorts. In strict-core LFO analyses, functional-change risk gradients were observed within cohort, but continuous three-domain scores matched or exceeded categorical LFO profiles for discrimination, model fit and decision-curve net benefit in 15 of 20 strict-core threshold comparisons. Secondary mortality Cox models showed high-burden class gradients, but proportional-hazards/time-drift flags in several cohorts and incomplete hard-endpoint harmonization kept mortality secondary. **Conclusions:** For midlife and older women’s health research, the defensible contribution is a harmonized descriptive audit and cautionary profile-stability analysis using applied-for, downloaded cohort data cleaned by the authors. Categorical profiles may support burden mapping and communication, but the current evidence does not support stable latent endotypes, transportable categorical risk tools, population prevalence claims or prediction superiority over continuous domain scores. Continuous domain scores should remain the preferred analytic scale for functional-change modelling in this package.

Keywords: aging, women’s health, multimorbidity, functional change, frailty, intrinsic capacity, harmonization

1 Background

Women’s health research has historically emphasized reproductive health, cancer, mental health and selected chronic conditions, but health after midlife also depends on how functional ability, cognitive reserve, affective symptoms and chronic disease burden accumulate together. A women’s health framing after midlife is therefore not limited to reproductive or cancer outcomes; it also includes capacity, disability, mental health and chronic-disease burden that shape independence and care needs. Functional ability, intrinsic capacity and multimorbidity are central to health-system planning for ageing populations [2–5]. Frailty phenotypes and deficit-accumulation indices summarize vulnerability [6–8], but they can compress clinically different domain patterns into a single severity continuum.

12 This study is intentionally scoped to women aged 50 years or older. This midlife-
13 and-older threshold captures the transition into later-life health-risk accumulation
14 while retaining comparability across the available ageing cohorts. Women have longer
15 survival, higher late-life disability burden and different patterns of affective symptoms,
16 multimorbidity and care needs [9]. The source cohorts did not provide a harmo-
17 nized cross-cohort set of reproductive, menopausal or gynaecologic exposures for the
18 present analysis; therefore the clinical question is not a sex-specific mechanism ques-
19 tion. Instead, it asks how multidomain functioning, mental-health and chronic-disease
20 burden should be described in women-focused ageing research when measurement
21 equivalence and profile stability are uncertain. Exploratory all-sex LFO severity-by-
22 sex interaction screens were fitted on a separately standardized all-sex scale; only
23 one of six modelled cohorts had a nominal interaction signal, so these analyses do
24 not support a consistent women-specific mechanism. The women-only design is there-
25 fore framed as a focused women’s-health population study, not as evidence that the
26 observed patterns are unique to women.

27 Recent multidomain trajectory work suggests that functional, cognitive, mood and
28 chronic-disease indicators can evolve jointly [10]. The unresolved clinical and epi-
29 demologic problem is whether categorical burden profiles remain interpretable after
30 harmonization risk, complete-case selection, endpoint coupling and model instability
31 are made explicit. We therefore focus on conservative description and claim calibration:
32 whether multidomain burden can be summarized without overstating transportable
33 latent subtypes or prediction performance.

34 **2 Methods**

35 **2.1 Cohorts and analytic population**

36 After registration or data-use application as required by each study, we downloaded
37 de-identified cohort datasets from CHARLS, ELSA, HRS, KLoSA, LASI, MHAS and

38 SHARE official data access systems [11–17]. The authors cleaned the local cohort files,
39 retained project-specific derived outputs and assembled the present analysis dataset.
40 Variables included harmonized ageing-study products where available [1] and locally
41 assembled domain inputs when no directly comparable cleaned variable was avail-
42 able. The analytic population was women aged 50 years or older at the cohort-specific
43 selected baseline or analysis wave; we describe this population as midlife and older
44 women rather than as a uniformly geriatric population. Sex coding was confirmed
45 from local Working Data Stata value labels and do-files as **ragender=0** for women
46 and **ragender=1** for men. Source-screen, complete-domain, descriptive-profile and
47 functional-association denominators were separated in line with observational-study
48 reporting principles [18]. Detailed cohort construction, harmonization and baseline
49 covariate metadata are provided in Additional file 1.

50 **2.2 Domain construction and harmonization review**

51 Four domains were constructed: functional limitation, cognitive burden, affec-
52 tive symptoms and cardiometabolic/chronic disease burden. Functional items were
53 anchored to activities of daily living and instrumental activities of daily living con-
54 structs where available [19, 20]. Affective symptoms used CES-D-family information
55 where available [21]; SHARE used EURO-D information [22]. Scores were standardized
56 within cohort and oriented so that higher values indicated worse burden. Retrospec-
57 tive harmonization review treated similar orientation as insufficient evidence for item
58 identity or transportable measurement equivalence [23, 24].

59 **2.3 Evidence tiers and descriptive profiles**

60 The strict-core descriptive construction tier comprised CHARLS, ELSA, HRS and
61 MHAS. SHARE was retained as construction-descriptive but validation-downgraded
62 sensitivity evidence, KLoSA as functional-bridge sensitivity and LASI as baseline-
63 only descriptive evidence. Cohort-specific Gaussian mixture models with two to five

64 classes were fit to the four domain scores as descriptive clustering tools [25]. Models
65 used full covariance matrices, `reg_covar=1e-6`, `max_iter=500`, `n_init=5` and random
66 state 20260601. The selected model used the lowest BIC among converged models
67 with minimum class size at least 5%, with class enumeration interpreted cautiously
68 [26, 27]. The final stability guardrail used 200 nonparametric bootstrap refits per
69 selected solution, covariance diagnostics and cross-algorithm agreement; these were
70 treated as main guardrails rather than supplemental technicalities. Detailed profile
71 dictionaries and stability outputs are provided in Additional file 2.

72 **2.4 Functional-change association and unresolved endpoint**

73 **guardrails**

74 The functional-change endpoint was deterioration of at least 0.5 SD from baseline.
75 Because functional change is a within-domain endpoint and the original four-domain
76 descriptive profiles include baseline function, this analysis cannot be interpreted as
77 independent hard-outcome validation. We therefore rebuilt exploratory LFO profiles
78 after omitting the baseline functional domain and compared categorical LFO pro-
79 files with continuous cognitive, affective and cardiometabolic/chronic scores. Models
80 adjusted for age, education, marital status, smoking and drinking. LFO model outputs
81 are provided in Additional file 3. We additionally fitted formal secondary all-cause
82 mortality Cox models, post-baseline hospitalization candidate models where cleaned
83 binary headers existed, and calibration/decision-curve guardrails; these secondary end-
84 point and prediction guardrail outputs are provided in Additional file 4. Exploratory
85 all-sex LFO severity-by-sex interaction models and survey-design audits are provided
86 in Additional file 5. These analyses were treated as secondary or sensitivity evidence
87 rather than harmonized primary validation.

88 3 Results

89 3.1 Denominator lock, baseline profile and complete-case 90 selection

91 The source screen included 79,938 women aged 50 years or older, and 76,293
92 had complete four-domain descriptive assignments. The strict-core construction set
93 comprised 29,058 women from CHARLS, ELSA, HRS and MHAS. The sensitiv-
94 ity/descriptive construction set comprised 47,235 women from SHARE, KLoSA and
95 LASI. The complete-domain 50–64-year stratum ranged from 45.4% in HRS to 67.8%
96 in CHARLS, supporting the midlife-and-older scope rather than a uniformly geriatric
97 interpretation. These counts and percentages are analytic-sample summaries rather
98 than survey-weighted population estimates. Complete-case selection was not neutral:
99 excluded participants were older than included complete-domain participants in every
100 cohort, including HRS (+10.4 years) and ELSA (+6.6 years). Table 1 and Figure 1 sep-
101 arate source, complete-domain, endpoint-availability and LFO model denominators.
102 Each downstream analysis used its own endpoint- and covariate-complete denomi-
103 nator, so LFO, mortality and descriptive-construction denominators should not be
104 interchanged.

Table 1 Baseline clinical profile and analytic denominators

Cohort	Tier/wave	Source/ complete N	LFO N/events/ follow-up	Baseline clinical profile	Excluded age delta
CHARLS	Strict- core; w1	6,878 / 6,019 (87.5%)	5,691; 1,766 (31.0%); 9.0 y	Age 61.9; 50– 64/65+ 67.8/32.2%; BMI 23.9; chronic 0.7; 2+ chronic 17.6%; smoke/drink 6.5/15.0%	+4.2 y
ELSA	Strict- core; w1	6,292 / 6,104 (97.0%)	5,153; 1,316 (25.5%); 12.0 y	Age 65.1; 50–64/65+ 51.6/48.4%; BMI unavailable; chronic 0.7; 2+ chronic 14.9%; smoke/drink 18.2/85.2%	+6.6 y
HRS	Strict- core; w5	11,005 / 10,202 (92.7%)	9,431; 3,546 (37.6%); 15.0 y	Age 67.2; 50– 64/65+ 45.4/54.6%; BMI 27.0; chronic 1.0; 2+ chronic 25.2%; smoke/drink 14.4/39.8%	+10.4 y
KLoSA	Bridge sensitivity; w3	4,344 / 4,081 (93.9%)	3,834; 1,144 (29.8%); 10.0 y	Age 66.0; 50– 64/65+ 47.1/52.9%; BMI 23.4; chronic 0.7; 2+ chronic 17.4%; smoke/drink 3.0/17.7%	+4.0 y
LASI	Baseline- only; no wave field	28,165 / 27,433 (97.4%)	Unavailable in current cleaned pass	Age 62.6; 50– 64/65+ 61.5/38.5%; BMI 23.1; chronic 0.6; 2+ chronic 13.5%; smoke/drink 3.7/2.6%	+6.5 y
MHAS	Strict- core; w1	7,440 / 6,733 (90.5%)	5,443; 1,437 (26.4%); 16.0 y	Age 62.4; 50– 64/65+ 63.8/36.2%; BMI 27.7; chronic 0.7; 2+ chronic 13.0%; smoke/drink 9.5/17.6%	+4.5 y
SHARE	Validation down; w1	15,814 / 15,721 (99.4%)	12,205; 1,525 (12.5%); 11.0 y	Age 64.8; 50– 64/65+ 53.9/46.1%; BMI 26.2; chronic 0.8; 2+ chronic 22.1%; smoke/drink 15.1/59.0%	+3.3 y

Source denominators are women aged 50 years or older in the selected baseline screen. Complete-domain denominators are four-domain descriptive assignments. The 50–64/65+ split and baseline clinical summaries are calculated among complete-domain participants unless unavailable. Follow-up years are nominal cohort intervals used for the functional-change endpoint. Endpoint availability is counted before the covariate-complete LFO modelling restrictions used in the LFO association analysis and Figure 1. Education, marital status, smoking, drinking and rural/region field availability is reported in Additional file 1.

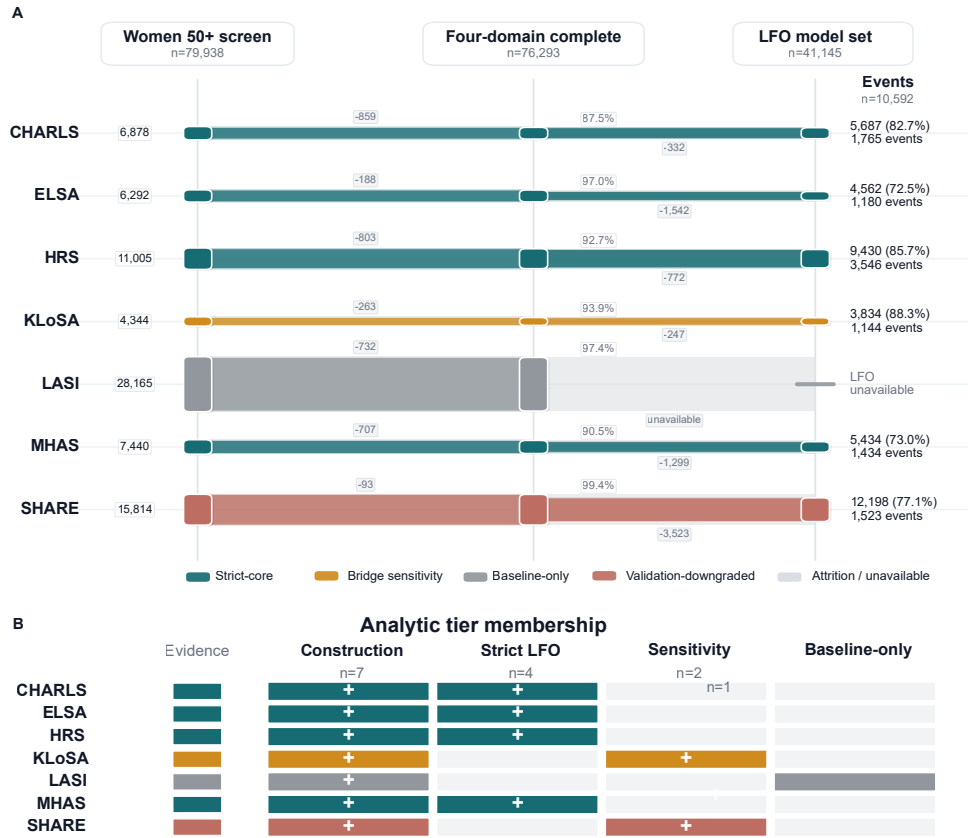


Fig. 1 Analytic denominator flow and evidence-tier membership. Panel A shows alluvial-style per-cohort attrition from women aged 50 years or older source screening to complete four-domain construction and covariate-complete LFO model denominators. These model-complete LFO denominators are intentionally smaller than the endpoint-available denominators in Table 1. Panel B summarizes analytic-tier membership for construction, strict-core LFO association, sensitivity and baseline-only descriptive roles. LASI remains baseline-only in the current cleaned-data pass.

3.2 Profile-stability guardrails

All selected full-covariance four-domain Gaussian mixture solutions triggered near-singular covariance diagnostics, with minimum covariance eigenvalues pinned at the regularization floor in at least one component per cohort. In the 200-refit bootstrap guardrail, all refits converged, but lower-tail agreement remained weak in several cohorts: p10 ARI was 0.35 in ELSA, 0.28 in HRS, 0.23 in KLoSA and 0.37 in SHARE.

111 Even cohorts with moderate-to-high median ARI were downgraded by near-singular
112 covariance and weak lower-tail bootstrap agreement. The profile labels are therefore
113 descriptive partitions for summarizing observed burden combinations, not stable latent
114 disease entities or transportable risk tools (Table 2; Figure 2). The strict-core profile-
115 pattern heatmap is retained as a supplementary descriptive display rather than as
116 primary validation evidence.

Table 2 Model-stability guardrails for selected four-domain profile solutions

Cohort	Evidence tier	k	Bootstrap ARI	Cross-method ARI	Max condition	Claim status
CHARLS	Strict-core	3	1.00 (1.00; 0.43)	0.26 (0.12-1.00)	1.20e+06	Descriptive only
ELSA	Strict-core	5	0.84 (0.35; 0.23)	0.25 (0.22-1.00)	1.35e+06	Descriptive only
HRS	Strict-core	5	0.76 (0.28; 0.27)	0.20 (0.17-0.92)	1.52e+06	Descriptive only
KLoSA	Bridge sensitivity	3	0.98 (0.23; 0.21)	0.62 (0.22-0.93)	1.36e+06	Bridge sensitivity
LASI	Baseline-only	3	0.99 (0.97; 0.16)	0.22 (0.09-0.96)	1.19e+06	No follow-up validation
MHAS	Strict-core	5	0.98 (0.84; 0.33)	0.17 (0.15-0.95)	2.53e+06	Descriptive only
SHARE	Validation-downgraded	4	0.37 (0.37; 0.36)	0.22 (0.19-0.99)	9.28e+05	Downgraded

ARI = adjusted Rand index. Bootstrap ARI is median (p10; minimum) across 200 non-parametric refits. Cross-method ARI is median (range) against diagonal/tied GMM, k-means, sampled hierarchical clustering and continuous-severity tertiles. All selected full-covariance solutions triggered near-singular covariance diagnostics; therefore these profiles are descriptive partitions and not stable latent subtypes.

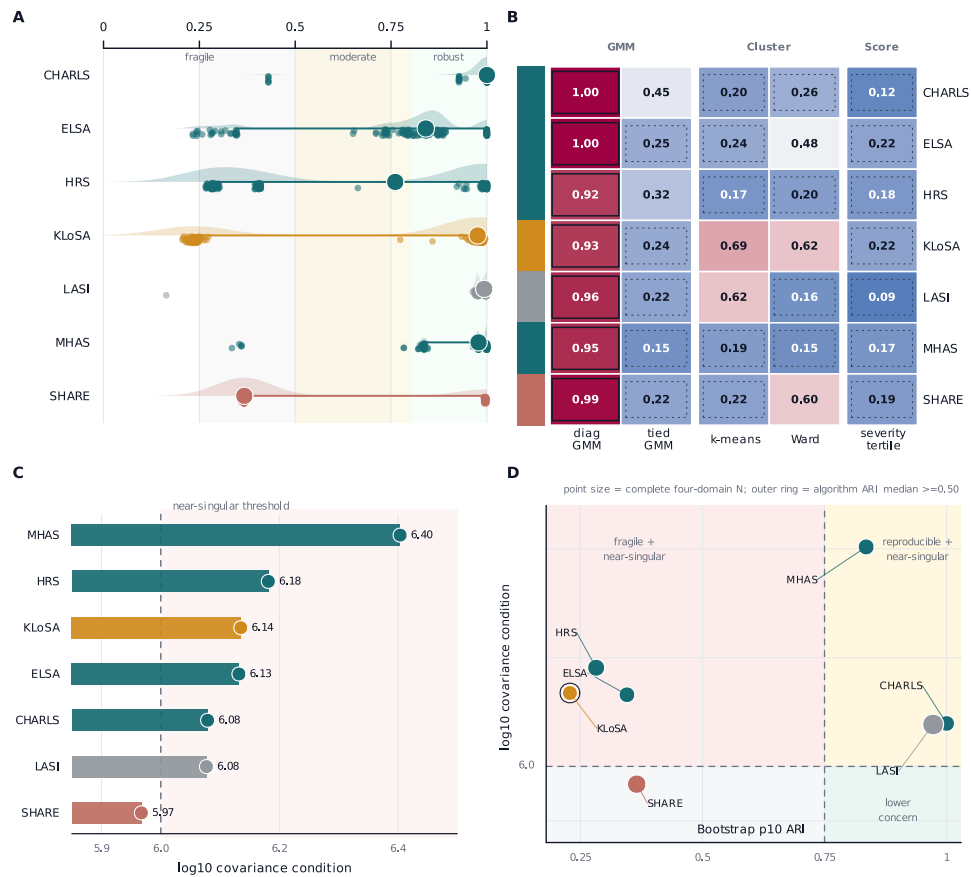


Fig. 2 Model-stability guardrails for selected descriptive profile partitions. Panel A summarizes 200-refit bootstrap ARI distributions, with higher values indicating greater refit stability. Panel B shows cross-method ARI against the selected full-covariance GMM across alternative clustering and continuous-severity comparators. Panel C ranks cohorts by log10 covariance condition number with the near-singular threshold shown as a reference line. Panel D maps bootstrap p10 ARI against log10 covariance condition to separate reproducible, fragile and near-singular descriptive solutions; it is a visualization of numerical guardrails, not a separate statistical model.

3.3 Exploratory LFO functional-change associations

Strict-core LFO functional-change analyses included 25,113 women and 7,925 functional deterioration events. Highest crude-risk LFO classes showed within-cohort functional-change gradients, although the ELSA highest-class contrast was not statistically conclusive. Continuous three-domain scores matched or outperformed categorical LFO profiles for discrimination and model fit in every strict-core cohort (Table 3;

Figure 3). Random-effects heterogeneity for the strict-core adjusted LFO risk ratios was considerable (pooled RR 1.40, 95% CI 1.19 to 1.64; I^2 80.2%; Q $p=0.002$; Additional file 3). With heterogeneity near 80%, the pooled estimate is a descriptive cross-cohort summary across non-identical instruments rather than a transportable effect estimate. These results describe within-cohort association with a functional change-score endpoint; they do not validate the four-domain descriptive profile patterns and do not demonstrate prediction superiority. SHARE and KLoSA are provided only as sensitivity rows because SHARE remained validation-downgraded and KLoSA used a functional bridge.

Table 3 Exploratory leave-functional-domain-out functional-change associations in strict-core cohorts

Cohort	LFO N/events	Risk contrast	Adjusted RR	Delta AUC	Reading
CHARLS	5,687/1,765 (31.0%)	C1 21.5% to C3 39.8%; RD 18.3 pp	1.63 (1.35 to 1.96)	-0.015	Continuous favoured
ELSA	4,562/1,180 (25.9%)	C1 21.1% to C5 42.2%; RD 21.1 pp	1.15 (0.96 to 1.37)	-0.007	Continuous favoured
HRS	9,430/3,546 (37.6%)	C1 28.4% to C4 51.8%; RD 23.4 pp	1.28 (1.16 to 1.41)	-0.008	Continuous favoured
MHAS	5,434/1,434 (26.4%)	C1 20.6% to C3 36.6%; RD 16.0 pp	1.61 (1.42 to 1.82)	-0.002	Continuous favoured

LFO = leave-functional-domain-out; RD = absolute risk difference between class 1 and the highest crude-risk LFO class. LFO classes were rebuilt without the baseline functional domain and are not the same objects as the four-domain descriptive profile patterns. Adjusted risk ratios use modified Poisson models with robust standard errors and adjustment for age, education, marital status, smoking and drinking. Delta AUC is profile AUC minus continuous-score AUC; negative values favour continuous three-domain scores. Full profile and continuous AUC values, SHARE/KLoSA sensitivity rows and class-level risks are provided in Additional file 3.

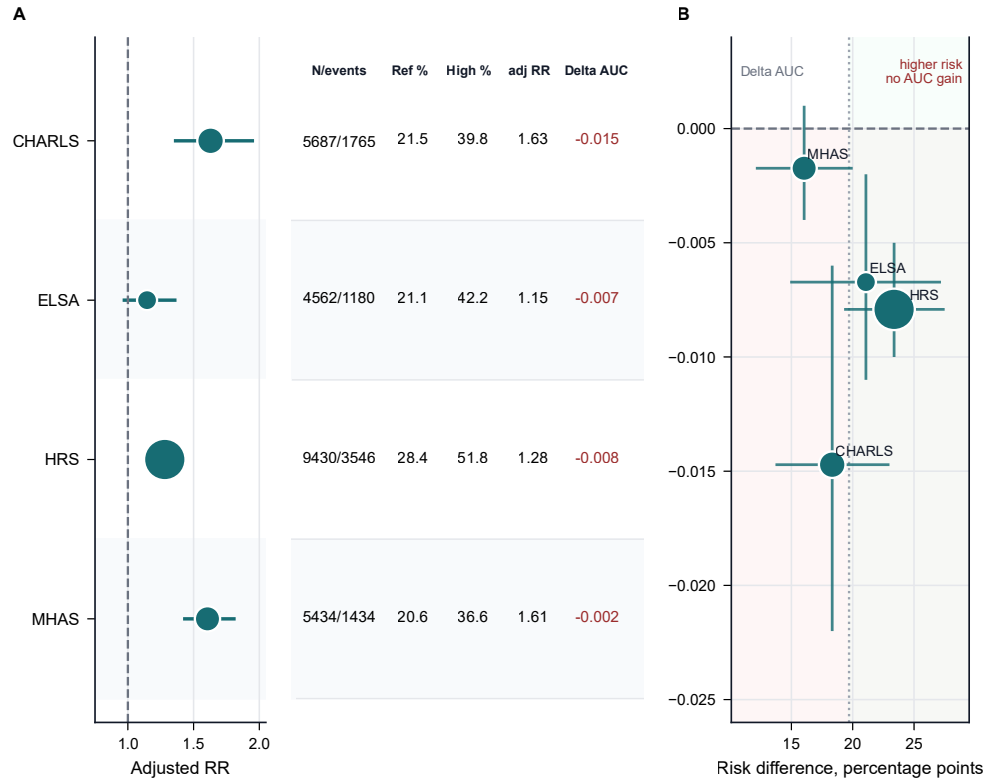


Fig. 3 Exploratory strict-core LFO functional-change associations. Panel A shows adjusted risk ratios comparing the highest crude-risk LFO class with the reference class, with point size scaled by model denominator and the adjacent table reporting N/events, reference-class risk, highest-class risk, adjusted RR and delta AUC. Panel B places cohorts in risk-difference versus delta-AUC space; horizontal intervals show uncertainty in crude risk difference and vertical intervals show 95% bootstrap intervals for delta AUC. Delta AUC is profile AUC minus continuous-score AUC, so negative values favour continuous three-domain scores.

3.4 Secondary mortality and prediction-guardrail analyses

Formal secondary all-cause mortality Cox models were fitted in six cohorts; LASI remained unavailable for mortality follow-up in the current cleaned pass. Highest-burden profile classes were associated with higher all-cause mortality in the modelled cohorts, but proportional-hazards or time-drift guardrail flags appeared in ELSA, HRS, KLoSA and SHARE. The ELSA death count was low relative to the nominal follow-up interval and may reflect incomplete mortality ascertainment in the current cleaned pass, so the ELSA hazard ratio should be read with particular caution.

140 Among strict-core cohorts, random-effects heterogeneity for the highest-class mortal-
 141 ity HR was considerable (pooled HR 2.37, 95% CI 1.99 to 2.82; I^2 78.7%; Q $p=0.003$;
 142 Additional file 4). With considerable heterogeneity and PH/time-drift flags in sev-
 143 eral cohorts, the pooled HR and flagged cohort-specific HRs are descriptive guardrail
 144 summaries rather than constant-hazard or transportable effect estimates. Mortality
 145 is therefore reported as a secondary non-circular endpoint guardrail rather than as
 146 primary validation evidence (Table 4). Decision-curve guardrails also supported the
 147 main cautionary interpretation: continuous three-domain scores had higher apparent
 148 in-sample net benefit than categorical LFO profiles in 15 of 20 strict-core threshold
 149 comparisons, without optimism correction. Calibration outputs are likewise reported
 150 only as apparent in-sample diagnostics and are not interpreted as validated calibration.

151 **3.5 Harmonization, hard-outcome and survey-design guardrails**

152 The harmonization matrix remained central to interpretation (Figure 4). Functional-
 153 domain strictness differed across cohorts: CHARLS used IADL-only information,
 154 HRS used ADL-only information, KLoSA used a bridge proxy and SHARE remained
 155 validation-downgraded. Cognitive batteries were cohort-specific, SHARE affective
 156 symptoms used EURO-D, and cardiometabolic/chronic disease indicators differed
 157 in lipid/cholesterol availability. Raw/codebook/dofile auditing identified candidate
 158 weight, primary sampling unit and strata triplets in HRS, KLoSA, LASI and SHARE,
 159 but no codebook-confirmed harmonized seven-cohort survey-design triplet was avail-
 160 able, so no survey-weighted prevalence claim is made. The all-sex interaction review
 161 found one nominal LFO severity-by-sex interaction among six modelled cohorts, which
 162 supports the focused women-only scope but not a consistent women-specific mech-
 163 anism. Hospitalization candidate models were feasible only where cleaned binary
 164 headers existed and remain header-only sensitivity evidence until codebook mapping is
 165 completed; institutionalization and care-dependence endpoints remained unavailable.

Table 4 Secondary all-cause mortality Cox guardrail models

Cohort	Tier	N/deaths; time	Highest class HR	PH/time flag	Reading
CHARLS	strict-core	5,868; 703 deaths (12.0%); 9.0 y	C3 HR 1.82 (1.54 to 2.16)	No flag	Secondary
ELSA	strict-core	4,639; 317 deaths (6.8%); 12.0 y	C5 HR 3.07 (2.15 to 4.39)	Flagged	Time-averaged
HRS	strict-core	10,043; 5,569 deaths (55.5%); 15.9 y	C5 HR 2.60 (2.34 to 2.89)	Flagged	Time-averaged
KLoSA	bridge sens.	3,990; 726 deaths (18.2%); 10.0 y	C3 HR 1.30 (1.09 to 1.56)	Flagged	Bridge sensitivity
LASI	baseline-only	unavailable	NA	NA	Not modelled
MHAS	strict-core	6,477; 2,231 deaths (34.4%); 17.0 y	C3 HR 2.39 (2.10 to 2.72)	No flag	Secondary
SHARE	validation-down	12,952; 3,198 deaths (24.7%); 11.0 y	C4 HR 2.70 (2.40 to 3.03)	Flagged	Time-averaged

These are formal secondary all-cause mortality Cox models using available death timing fields after local data cleaning. They are reported as non-circular endpoint guardrails, not as primary validation, because proportional-hazards or time-drift flags were present in several cohorts, survey-design variables were not applied, and LASI mortality follow-up was unavailable in the current cleaned pass. The ELSA death count appeared low relative to nominal follow-up and may reflect incomplete mortality ascertainment in the current cleaned pass; the ELSA HR should therefore be interpreted with particular caution. HRs in flagged cohorts should be read as conventional time-averaged guardrail summaries rather than constant-hazard effects. Technical flag coding and AIC guardrail details are provided in Additional file 4.

166 These guardrails inform the clinical interpretation and claim boundaries summarized
167 in Box 1 in the Discussion. The full workflow and supplementary profile displays are
168 embedded after the references as Supplementary Figures S1-S5.

169 4 Discussion

170 This manuscript is deliberately scoped as a harmonized descriptive audit and cau-
171 tionary profile-stability analysis for midlife and older women’s health research. The
172 seven-cohort comparison shows that multidomain burden among women aged 50 years
173 or older can be summarized and audited, but the selected categorical partitions are

CHARLS	Partial 99.1% IADL only	Cohort-specific 87.6% cognitive score	Primary 92.8% CES-D family	Primary 99.3% chronic count
ELSA	Primary 98.5% ADL/IADL	Cohort-specific 97.5% cognitive score	Primary 97.1% CES-D family	Primary 99.9% chronic count
HRS	Partial 99.8% ADL only	Cohort-specific 92.8% cognitive score	Primary 92.8% CES-D family	Primary 100.0% chronic count
KLoSA	Bridge 100.0% grip/falls proxy	Cohort-specific 94.7% cognitive score	Primary 99.1% CES-D family	Primary 100.0% chronic count
LASI	Primary 99.7% ADL/IADL	Cohort-specific 98.8% cognitive score	Primary 97.4% CES-D family	Primary 99.8% chronic count
MHAS	Primary 93.9% ADL/IADL	Cohort-specific 92.7% cognitive score	Primary 93.7% CES-D family	Primary 97.9% chronic count
SHARE	Primary 100.0% ADL/IADL	Cohort-specific 99.5% cognitive score	Alternate instrument 100.0% EURO-D	Primary 99.5% chronic count
	Functional	Cognitive	Affective	Cardiometabolic/ chronic
	Primary source	Cohort-specific / partial cognition	Partial or alternate instrument	Bridge proxy

Fig. 4 Cohort-domain harmonization risk matrix. Cells summarize source tier, non-missing percentage and the principal construct used for each burden domain. The matrix is measurement-risk evidence rather than a biological result.

not robust enough to be interpreted as stable endotypes or transportable risk tools.

The main positive contribution is transparent measurement and denominator mapping across several major ageing cohorts; the main cautionary finding is that categorical profile solutions are numerically fragile and do not improve on continuous domain scores for the current functional-change endpoint.

Box 1. Clinical interpretation and claim boundaries for women’s health research

Evidence anchor	Interpretation	Boundary
Table 1 and Figure 1	A large midlife-and-older women analytic sample can be described with explicit denominator attrition and tier membership. Categorical profiles can summarize multidomain burden patterns and support communication.	Counts are analytic-sample summaries, not survey-weighted population prevalence estimates. Full-covariance GMM solutions were numerically fragile and should not be treated as stable latent endotypes or transportable risk tools.
Table 2 and Figure 2		
Table 3, Figure 3 and Additional file 3	Continuous LFO cognitive, affective and cardiometabolic/chronic scores matched or exceeded categorical profiles for functional-change modelling.	The analysis does not show categorical prediction superiority; continuous domain scores remain the preferred analytic scale in this package.
Table 4, Figure 4 and Additional files 4-5	Secondary mortality gradients and harmonization/survey-design audits provide clinically relevant guardrails for further work.	Mortality remains secondary, hard-endpoint harmonization is incomplete and the all-sex interaction screen does not establish a consistent women-specific mechanism.

LFO = leave-functional-domain-out. This box is a discussion-level claim-calibration summary, not a separate statistical model.

The functional-change analysis should be interpreted conservatively. The LFO approach reduces predictor-side endpoint coupling by removing baseline function from the profile-construction step, but the outcome remains a within-domain change score rather than an independent hard clinical endpoint. The observed risk gradients therefore support exploratory within-cohort association, not validation of the four-domain descriptive patterns. This limitation is especially important because the formal secondary mortality analyses show that clinically more independent endpoints are partly feasible, while also demonstrating why mortality must remain secondary in this package because several cohorts triggered proportional-hazards or time-drift guardrails.

The women-only scope is also conservative. The analysis addresses midlife and older women as a clinically important population with high late-life disability burden, but

192 it does not establish sex-specific mechanisms. Exploratory all-sex LFO severity-by-sex
193 interaction models were fitted on a separately standardized all-sex scale and showed
194 no consistent cross-cohort interaction signal; only CHARLS had a nominal interac-
195 tion. These models were not used for women-only profile construction or outcome
196 modelling. A stronger mechanistic women’s-health paper would require a prespecified
197 all-sex design, formal sex-interaction estimands and harmonized women-specific expo-
198 sures. The present implication is narrower: women’s health studies after midlife should
199 report multidomain burden with transparent denominator, harmonization and stabil-
200 ity guardrails, and should retain continuous domain scores when modelling functional
201 change.

202 **5 Strengths and limitations**

203 Strengths include the seven-cohort scope, codebook-confirmed sex coding, explicit
204 denominator locking, separation of strict-core primary evidence from sensitivity
205 tiers, baseline clinical characterization, included-versus-excluded missingness auditing,
206 item-level harmonization review, LFO functional-change modelling, formal secondary
207 mortality modelling, all-sex interaction screens and 200-refit algorithm-stability
208 diagnostics. Limitations remain substantial. Domain measures were harmonized by ori-
209 entation and within-cohort standardization but were not instrument-identical across
210 cohorts; this is particularly important for Table 3 because functional instruments
211 differed by cohort. Complete-case selection likely removed older and higher-burden
212 participants. Survey weights, strata and primary sampling units were screened using
213 cleaned headers, Stata metadata and dofile/codebook text, but no common codebook-
214 confirmed seven-cohort design triplet was available. LASI lacked a follow-up validation
215 denominator, SHARE was validation-downgraded and KLoSA used a functional

216 bridge. Mortality was modelled only as secondary evidence; hospitalization mod-
217 els remain header-only candidate analyses; institutionalization and care-dependence
218 endpoints remain unavailable. Box 1 consolidates the resulting claim boundaries.

219 **6 Conclusions**

220 Across seven international ageing cohorts, multidomain burden among midlife and
221 older women can be described, tiered and audited. The conclusion is cautionary:
222 categorical profiles are descriptive partitions with important measurement, stabil-
223 ity and endpoint-coupling caveats. For functional-change modelling in this package,
224 continuous domain scores remain the preferred analytic scale.

225 **Abbreviations**

226 ADL, activities of daily living; AIC, Akaike information criterion; ARI, adjusted Rand
227 index; BIC, Bayesian information criterion; CES-D, Center for Epidemiologic Studies
228 Depression Scale; CHARLS, China Health and Retirement Longitudinal Study; ELSA,
229 English Longitudinal Study of Ageing; GMM, Gaussian mixture model; HRS, Health
230 and Retirement Study; IADL, instrumental activities of daily living; KLoSA, Korean
231 Longitudinal Study of Aging; LASI, Longitudinal Aging Study in India; LFO, leave-
232 functional-domain-out; MHAS, Mexican Health and Aging Study; SHARE, Survey of
233 Health, Ageing and Retirement in Europe.

234 **Declarations**

235 **Ethics approval and consent to participate**

236 This study used approved, downloaded, de-identified cohort datasets from CHARLS,
237 ELSA, HRS, KLoSA, LASI, MHAS and SHARE after registration or data-use appli-
238 cation as required by each cohort. Each source cohort obtained ethics approval and

239 participant consent according to its protocol. The authors cleaned and harmonized the
240 analysis files locally. The present analysis did not involve direct participant contact.

241 **Consent for publication**

242 Not applicable.

243 **Availability of data and materials**

244 The source cohort datasets are available through their respective official data portals
245 or data access systems subject to registration, application approval and data-use con-
246 ditions. The authors downloaded, cleaned and harmonized the approved de-identified
247 cohort files for this analysis. Source and derived participant-level data cannot be redis-
248 tributed if doing so would violate source cohort data-use agreements. Analysis code,
249 harmonization metadata and aggregate non-identifying outputs used to reproduce the
250 manuscript tables and figures are archived at Zenodo: DOI [10.5281/zenodo.20589522](https://doi.org/10.5281/zenodo.20589522),
251 with the development version available at GitHub: [https://github.com/Luciky-Leo/](https://github.com/Luciky-Leo/older-women-multidomain-burden)
252 [older-women-multidomain-burden](https://github.com/Luciky-Leo/older-women-multidomain-burden).

253 **Competing interests**

254 The authors declare that they have no competing interests.

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257 **Authors' contributions**

258 FL, JC and JS contributed equally to this work and share first authorship. FL con-
259 tributed to study design, data cleaning, harmonization, statistical analysis, table and
260 figure assembly and drafting of the manuscript. JC contributed to clinical interpre-
261 tation of functional-change and secondary endpoint findings, assessment of clinical

262 relevance and critical revision of the manuscript. JS contributed to the women's-health
263 framing, obstetrics-and-gynaecology clinical interpretation, literature interpretation
264 and critical revision of the manuscript. LL contributed to clinical supervision, inter-
265 pretation of women's-health implications and critical revision of the manuscript. RG
266 conceived and supervised the study, led the statistical strategy with FL, verified the
267 analyses and finalized the manuscript. RG is the primary corresponding author and
268 LL is a co-corresponding author. All authors read and approved the final manuscript.

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271 **Use of artificial intelligence tools**

272 Large language model tools were used for drafting support, code-execution log sum-
273 marization and language editing. No artificial intelligence tool was listed as an author.
274 The authors verified all analyses, references, figures and manuscript content and take
275 full responsibility for the final work.

276 **Additional files**

277 Additional file 1. File name: `additional_file_1_harmonization_and_cohort_`
278 `construction.xlsx`. File format: XLSX. Title: Cohort construction, harmonization
279 and baseline covariate metadata. Description: Workbook containing item-level har-
280 monization crosswalks, cohort tier lock, harmonization risk matrix, midlife/older age
281 distribution and baseline clinical/design/covariate availability tables.

282 Additional file 2. File name: `additional_file_2_profile_stability_and_`
283 `descriptive_profiles.xlsx`. File format: XLSX. Title: Descriptive profile and
284 model-stability guardrails. Description: Workbook containing GMM stability sum-
285 maries, 200-refit bootstrap robustness tables, covariance diagnostics, selected class

286 dictionary, descriptive profile summaries, profile-stability guardrail tables, algorithm
287 robustness sensitivity and sensitivity-cohort profile support.

288 Additional file 3. File name: `additional_file_3_lfo_functional_change_`
289 `associations.xlsx`. File format: XLSX. Title: Leave-functional-domain-out
290 functional-change association outputs. Description: Workbook containing decoupled
291 validation comparisons, functional endpoint leakage audit, strict-core LFO association
292 table, sensitivity rows, class-level risks, AUC bootstrap intervals and cross-cohort
293 heterogeneity summaries for functional-change analyses.

294 Additional file 4. File name: `additional_file_4_secondary_endpoints_and_`
295 `prediction_guardrails.xlsx`. File format: XLSX. Title: Secondary endpoint and
296 prediction guardrail outputs. Description: Workbook containing formal mortality
297 guardrail models, hospitalization candidate models/status tables, hard-endpoint fea-
298 sibility matrix, calibration metrics, decision-curve summaries, secondary endpoint
299 prediction guardrails and cross-cohort mortality heterogeneity summaries.

300 Additional file 5. File name: `additional_file_5_survey_design_and_sex_`
301 `comparator_audits.xlsx`. File format: XLSX. Title: Survey-design and sex-
302 comparator audits. Description: Workbook containing survey-design variable
303 and raw/codebook audits, survey triplet status, baseline male-comparator sum-
304 maries/contrasts, all-sex LFO sex-interaction terms and a reviewer-risk summary
305 sheet.

306

307 Supplementary Figures S1-S5 are embedded after the references in this manuscript
308 file and are not separate tabular Additional files.

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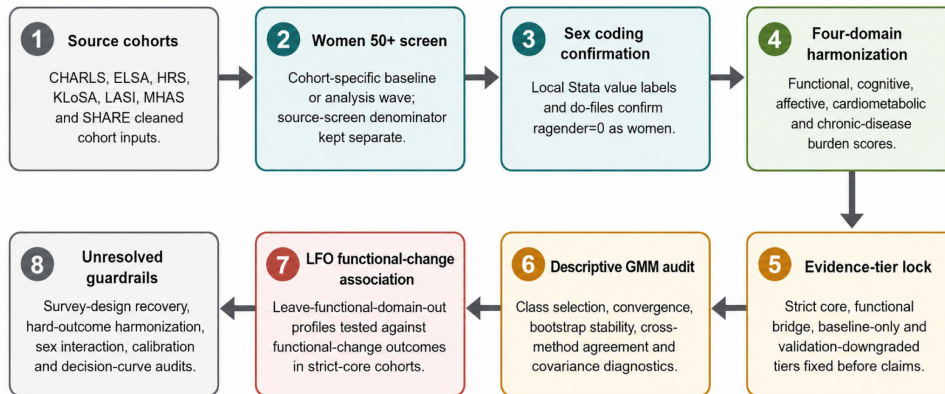
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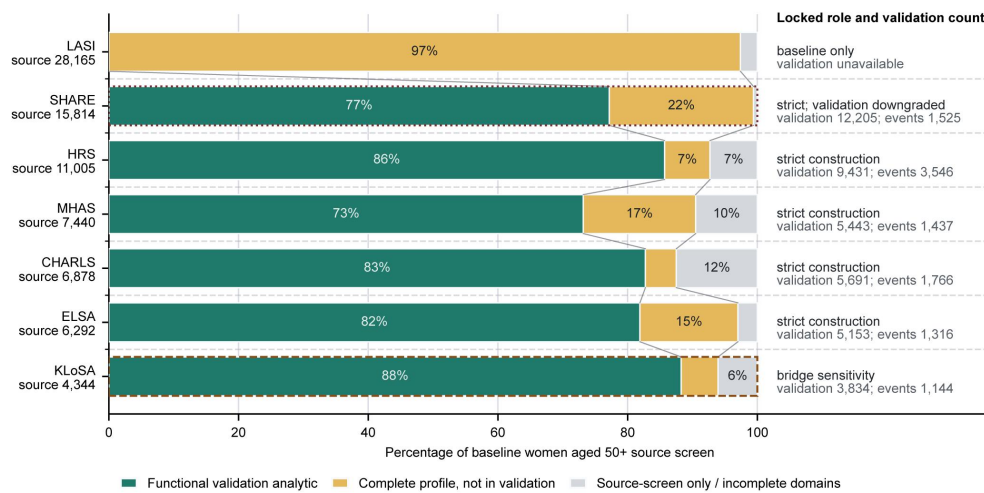
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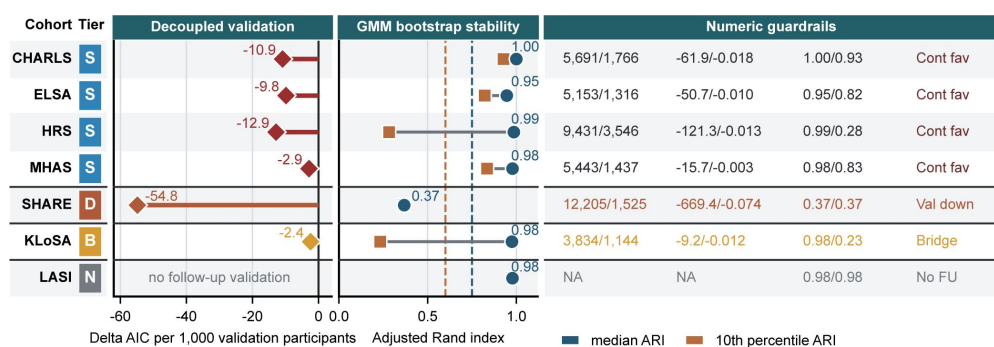
405 **Supplementary figures**



Supplementary Figure S1. Workflow schematic showing the analysis chain from source cohorts through women aged 50 years or older screening, sex-coding confirmation, four-domain harmonization, evidence-tier locking, descriptive GMM audit, leave-functional-domain-out functional-change association and unresolved guardrail audits.



Supplementary Figure S2. Cohort denominator and validation dashboard showing source women aged 50 years or older, complete-domain analytic proportions, LFO validation availability and evidence-tier labels.



Supplementary Figure S5. Decoupled validation, GMM bootstrap stability and numerical guardrail dashboard retained as supplementary context.